The documentation and process conversion measures necessary to comply with this document shall be completed by 9 February 2025.

INCH-POUND

MIL-PRF-19500/535G
w/AMENDMENT 1
9 September 2024
SUPERSEDING
MIL-PRF-19500/535G
5 June 2019

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, PNP, SILICON, POWER, TYPES 2N5003 AND 2N5005,
ENCAPSULATED (CASE MOUNT PACKAGE) AND UNENCAPSULATED,
QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for PNP, silicon, power transistors for use in high-speed power-switching applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type and two levels of product assurance (JANHC and JANKC) are provided for unencapsulated devices as specified in MIL-PRF-19500.

1.2 Physical dimensions.

1.2.1 Package outlines. The device package outline is a TO-210AA (formerly TO-59) in accordance with figure 1 for all encapsulated device types.

1.2.2 Unencapsulated die. The dimensions and topography for JANHC and JANKC unencapsulated die are in accordance with figure 2.

1.3 Maximum ratings unless otherwise specified T_A = +25°C.

P_T (1) T_A = 25°C	P_T (2) T_C = 25°C	$R_{\theta JA}$	$R_{\theta JC}$ (3)	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_C (4)	Reverse pulse energy (5)	T_{stg} and T_J
W	W	°C/W	°C/W	V dc	V dc	V dc	A dc	A dc	mJ	°C
2	58	88	3	-100	-80	-5.5	-5	-10	15	-65 to +200

(1) Derate linearly 11.4 mW/°C for T_A > +25°C.
(2) For derating see figure 3.
(3) For thermal impedance see figure 4.
(4) This value applies for P_W ≤ 8.3 ms, duty cycle ≤ 1 percent.
(5) This rating is based on the capability of the transistors to operate safely in the unclamped inductive load energy test circuit figure 5 herein.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

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1.4 Primary electrical characteristics. Unless otherwise specified T_C = +25°C.

Limits	h_{FE2} (1) V_{CE} = -5 V I_C = -2.5 A		$\|h_{fe}\|$ V_{CE} = -5 V I_C = -500 mA dc f = 10 MHz		$V_{BE(sat)2}$ (1) I_C = -5 A dc I_B = -500 mA dc	$V_{CE(sat)2}$ (1) I_C = -5 A dc I_B = -500 mA dc	C_{obo} V_{CB} = -10 V dc I_E = 0 f = 1 MHz
	2N5003	2N5005	2N5003	2N5005			
					V dc	V dc	pF
Min	30	70	6	7			
Max	90	200			-2.2	-1.5	250

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.2 JAN certification mark and quality level for unencapsulated die. The quality level designators for unencapsulated die that are applicable for this specification sheet are "JANHC" and "JANKC".

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

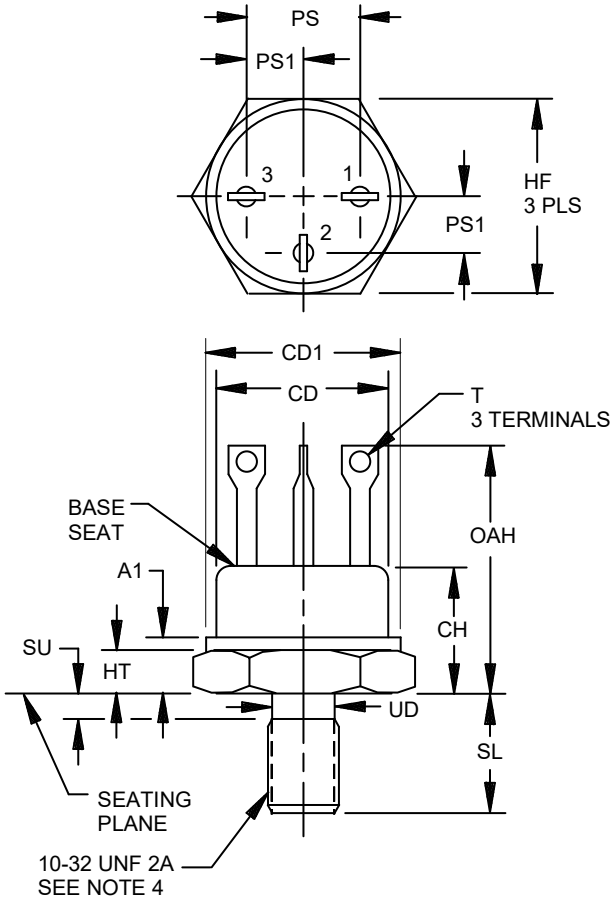
1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "5003" and "5005".

1.5.4 Die identifiers for unencapsulated die. The manufacturer and critical interface die identifiers that are applicable for this specification sheet is "A" (see figure 2 and 6.4.2).

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

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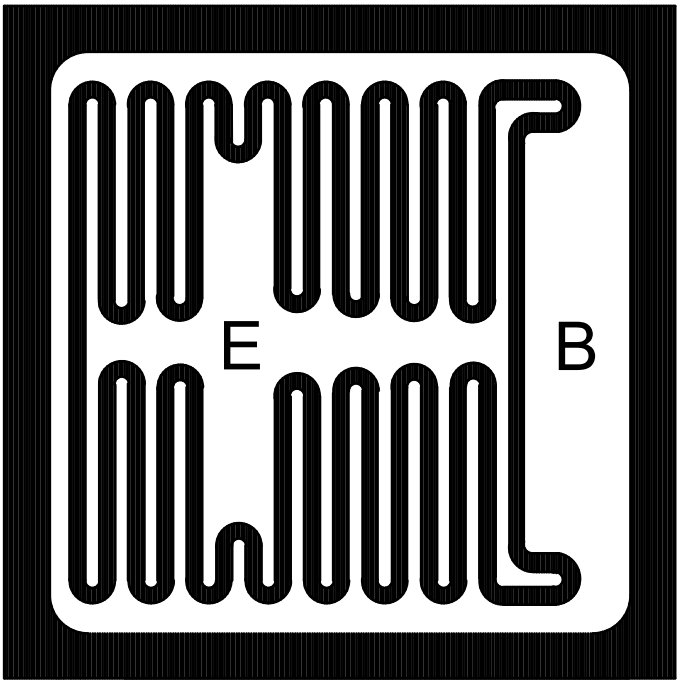


LTR	Dimension				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A1		.250		6.35	
CD	.330	.360	8.38	9.14	
CD1	.370	.437	9.40	11.10	
CH	.320	.468	8.13	11.89	
HF	.424	.437	10.77	11.10	
HT	.090	.150	2.29	3.81	
OAH	.575	.763	14.61	19.38	5
PS	.185	.215	4.70	5.46	6, 7, 8
PS1	.090	.110	2.29	2.79	6, 7, 8
SL	.400	.455	10.16	11.56	
SU		.078		1.98	9
T	.040	.065	1.02	1.65	
UD	.155	.189	3.94	4.80	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
* 2. Terminal 1 is emitter, terminal 2 is base, and terminal 3 is collector.
3. The case temperature may be measured anywhere on the seating plane within .125 inch (3.18 mm) of the stud.
4. Threads shall be 10-32 UNF-2A in accordance with FED-STD-H28. Maximum pitch diameter of plated threads shall be basic pitch diameter .1697 inch (4.31 mm).
5. All three terminals.
6. The angular orientation of the terminals in relation to the hex flats is not controlled.
7. This dimension applies to the location of the center line of the terminals.
8. Terminal spacing measured at the base seat only.
9. Maximum unthreaded dimension.

* FIGURE 1. Physical dimensions of transistor types 2N5003 and 2N5005 (TO-210AA, T6-B, similar to TO-59).

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NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
3. The physical characteristics of the die are as follow:
   a. Die size: .122 inch (0.31 mm) x .122 inch (0.31 mm).
   b. Die thickness: .008 (0.20 mm) to .012 (0.30 mm).
   c. Base pad (B): .015 (0.38 mm) x .0072 (.183 mm).
   d. Emitter pad (E): .015 (0.38 mm) x .0060 (.152 mm).
   e. Collector. Backside.
   f. Back metal: Gold 2,500 Å minimum, 3,000 Å nominal.
   g. Top metal: Aluminum, 40,000 Å minimum, 50,000 Å nominal.

FIGURE 2. Physical dimensions of JANHCA and JANKCA (A-version) die.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 2.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics shall be as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

d. Element evaluation (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening for quality levels JANTX and JANTXV and JANS. Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS level	Measurements for JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{CES1} and h_{FE2}	Not applicable
11	I_{CES1} and h_{FE2} ΔI_{CES1} = 100 percent or -100 nA, whichever is greater; Δh_{FE2} = ±20 percent.	I_{CES1} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein: ΔI_{CES1} = +100 percent of initial value or -100 nA, whichever is greater Δh_{FE2} = ±20 percent.	Subgroup 2 of table I herein: ΔI_{CES1} = +100 percent of initial value or -100 nA, whichever is greater Δh_{FE2} = ±20 percent.

(1) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions shall be as follows: V_{CB} = -10 to -30 V dc. Power shall be applied to achieve T_J = +175°C minimum using a minimum P_D = 75 percent of P_T maximum, T_A ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for quality levels JANTX and JANTXV. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval. NOTE: No heat sink or forced air cooling on the device shall be permitted.

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4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD}, t_{SW} (V_C and V_H where appropriate). See table II, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If the alternate procedure for screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of table I, subgroups 1 and 2 herein. The inspections of table E-VIB, subgroup 1 are not required to be performed again if group B has already been satisfied in accordance with 4.4.2.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with, table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX, and JANTXV.

4.4.2.1 Quality level JANS. Group B inspection for quality level JANS shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA of MIL-PRF-19500 and herein. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500. Delta electrical measurements shall be in accordance with the steps of 4.7 herein.

Subgroup	Method	Condition
B4	1037	V_{CB} = -10 V dc, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
B5	1026	V_{CB} = -10 V dc; P_D ≥ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table E-VIA, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve a T_J = +225°C minimum.
B6	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and in accordance with thermal impedance curves.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV. Group B inspection for quality levels JAN, JANTX, and JANTXV shall be conducted in accordance with the conditions specified in table E-VIC (small die flow) of MIL-PRF-19500 and herein.

Step	Method	Condition
1	1026	Steady-state life: Test condition B, 1,000 hours minimum, V_{CB} = -10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C.

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4.4.2.2.1 Group B sample selection. Samples selected for small die flow group B inspection shall be in accordance with all of the following requirements:

a. Samples shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. See MIL-PRF-19500.

b. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test may be selected prior to the application of final lead finish.

c. Separate samples may be used for each step.

4.4.2.2.2 Failures. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. All catastrophic failures during small die flow group B inspection shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein. Delta electrical measurements shall be in accordance with the applicable steps of 4.7 herein.

4.4.3.1 Quality level JANS.

Subgroup	Method	Condition
C2	2036	Tension: Test condition A, weight = 7 pounds ±5 ounces (3.2 ±0.16 kg), application time = 15 seconds; Terminal torque: Test condition D1, torque = 6 inch-ounce (42.4 mN-m), application time = 15 seconds; Stud torque: Test condition D2, torque = 15 inch-pounds (1.7 N-m), application time = 15 seconds.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and in accordance with thermal impedance curves.
C6	1026	1,000 hours at V_{CB} = -10 V dc; power shall be applied to achieve T_J = +175°C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3, n = 45, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Quality levels JAN, JANTX and JANTXV.

Subgroup	Method	Condition
C2	2036	Test condition A, weight = 7 pounds, ±5 ounces (3.2 ±0.16 kg), application time = 15 seconds; Test condition D1, torque = 6 inch-ounce (42.4 mN-m), application time = 15 seconds; Test condition D2, torque = 15 in-lbs (1.7 N-m), application time = 15 seconds.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3), and in accordance with thermal impedance curves.
C6		Not applicable.

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4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be selected prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta electrical measurements shall be in accordance with the applicable steps of 4.7.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified herein all inspections shall be conducted at a case temperature (T_C) of +25°C.

4.6 Element evaluation of unencapsulated die (quality levels JANHC and JANKC). The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. For subgroup 4, the burn-in duration for the JANKC level shall follow the JANS requirements and the JANHC level shall follow the JANTX requirements.

4.7 Delta measurements. Delta measurements shall be as specified below: (1) (2) (3) (4) (5)

Steps	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to emitter cutoff current	3041	Condition C, V_{CE} = -60 V dc	ΔI_{CES1}	100 percent of initial value or -100 nA, whichever is greater.		
2.	Forward-current transfer ratio	3076	I_C = -2.5 V dc, V_{CE} = -5 V dc pulsed (see 4.5.1)	Δh_{FE2}	±20 percent change from initial reading.		

(1) See MIL-PRF-19500 for sampling plan.
(2) Devices which exceed the table I limits for this test shall not be accepted.
(3) The delta electrical measurements for group B for quality level JANS are as follows:
  a. Subgroup 4, steps 1 and 2.
  b. Subgroup 5, steps 1 and 2.
(4) The delta electrical measurements for group B for quality levels JAN, JANTX and JANTXV) are as follows:
  a. Step 1, steps 1 and 2.
  b. Step 3, steps 1 and 2.
(5) The delta electrical measurements for group C for all quality levels are as follows: Subgroup 6, steps 1 and 2.
(6) The delta electrical measurements for group E for all quality levels are as follows: Subgroups 1 and 2, steps 1 and 2.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071					
Solderability 3/ 4/	2026					
Resistance to solvents 3/ 4/ 5/	1022					
Salt atmosphere (corrosion) 4/ 6/	1041	n = 6 devices, c = 0				
Temperature cycling 3/ 4/	1051	Test condition C, 25 cycles n = 22 devices, c = 0				
Hermetic seal 4/ 7/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hours or T_A = +300°C at t = 2 hours n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance 8/	3131	See 4.3.2	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = -100mA, I_B = 0, pulsed (see 4.5.1)	$V_{BR(CEO)}$	-80		V dc
Collector to emitter cutoff current	3041	Bias condition C, V_{BE} = 0, V_{CE} = -60 V dc	I_{CES1}		-1.0	µA dc
Collector to emitter cutoff current	3041	Bias condition C, V_{BE} = 0, V_{CE} = -100 V dc	I_{CES2}		-1.0	mA dc
Collector to emitter cutoff current	3041	Bias condition D, I_B = 0, V_{CE} = -40 V dc	I_{CEO}		-50	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection. - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Emitter to base cutoff current	3061	Bias condition C, I_C = 0, V_{EB} = -4 V dc	I_{EBO1}		-1.0	µA dc
Emitter to base cutoff current	3061	Bias condition D, I_C = 0, V_{EB} = -5.5 V dc	I_{EBO2}		-1.0	mA dc
Forward - current transfer ratio 2N5003 2N5005	3076	V_{CE} = -5 V dc, I_C = -50 mA dc	h_{FE1}	 20 50		
Forward - current transfer ratio 2N5003 2N5005	3076	V_{CE} = -5 V dc, I_C = -2.5 A dc, pulsed (see 4.5.1)	h_{FE2}	 30 70	 90 200	
Forward - current transfer ratio 2N5003 2N5005	3076	V_{CE} = -5 V dc, I_C = -5 A dc, pulsed (see 4.5.1)	h_{FE3}	 20 40		
Base-emitter voltage (nonsaturated)	3066	Test condition B, V_{CE} = -5 V dc, I_C = -2.5 A dc, pulsed (see 4.5.1)	$V_{BE(on)}$		-1.45	V dc
Base-emitter saturation voltage	3066	Test condition B, I_C = -2.5 A dc, I_B = -250 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		-1.45	V dc
Base-emitter saturation voltage	3066	Test condition B, I_C = -5 A dc, I_B = -500 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)2}$		-2.2	V dc
Collector-emitter saturation voltage	3071	I_C = -2.5 A dc, I_B = -250 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		-0.75	V dc
Collector-emitter saturation voltage	3071	I_C = -5 A dc, I_B = -500 mA dc pulsed (see 4.5.1)	$V_{CE(sat)2}$		-1.5	V dc
Subgroup 3						
High-temperature operation:		T_C = +150°C				
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = -60 V dc, V_{BE} = 0.	I_{CES}		-500	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection. - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 3						
Low-temperature operation:		T_C = -55°C				
Forward-current transfer ratio	3076	V_{CE} = -5 V dc, I_C = -2.5 A dc, pulsed (see 4.5.1)	h_{FE4}			
2N5003				15		
2N5005				25		
Common-emitter, small-signal, short-circuit, forward-current transfer ratio	3206	V_{CE} = -5 V dc, I_C = -100 mA dc, f = 1 kHz	h_{fe}			
2N5003				20		
2N5005				50		
Magnitude of common-emitter, small-signal short-circuit, forward-current transfer ratio	3306	V_{CE} = -5 V dc, I_C = -500 mA dc f = 10 MHz	$\|h_{fe}\|$			
2N5003				6		
2N5005				7		
Open-circuit output capacitance	3236	V_{CB} = -10 V dc, I_E = 0, f = 1 MHz	C_{obo}		250	pF
Switching time		I_C = -5 A dc, I_{B1} = -500 mA dc I_{B2} = -500 mA dc $V_{BE(off)}$ = -3.7 V R_L = 6Ω, (see figure 6)	t_{on} t_s t_f t_{off}		0.5 1.4 0.5 1.5	µs µs µs µs
Subgroup 5						
Safe operating area (dc)	3051	Pre-pulse condition for each test: V_{CE} = 0, I_C = 0, T_C = +25°C Pulse condition for each test t_p = 1 sec. 1 cycle T_C = +25°C, (see figure 7)				
Test #1		V_{CE} = -12 V dc, I_C = -5 A dc				
Test #2		V_{CE} = -32 V dc, I_C = -1.7 A dc				
Test #3		V_{CE} = -80 V dc, I_C = -100 mA dc				

See footnotes at end of table.

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TABLE I. Group A inspection. - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued.						
Safe operating area (unclamped inductive)	3053	Condition C, T_C = +25°C, R_{BB1} = 10Ω, R_{BB2} = 100Ω, L = 0.3 mH, RL = 0.1Ω, V_{CC} = -10 V dc V_{BB1} = -10 V dc, V_{BB2} = -4 V dc I_{CM} = -10 A dc (see figure 5)				
End-point electrical measurements		See 4.5.3 and subgroup 2 of table I.				
Subgroups 6 and 7						
Not applicable						

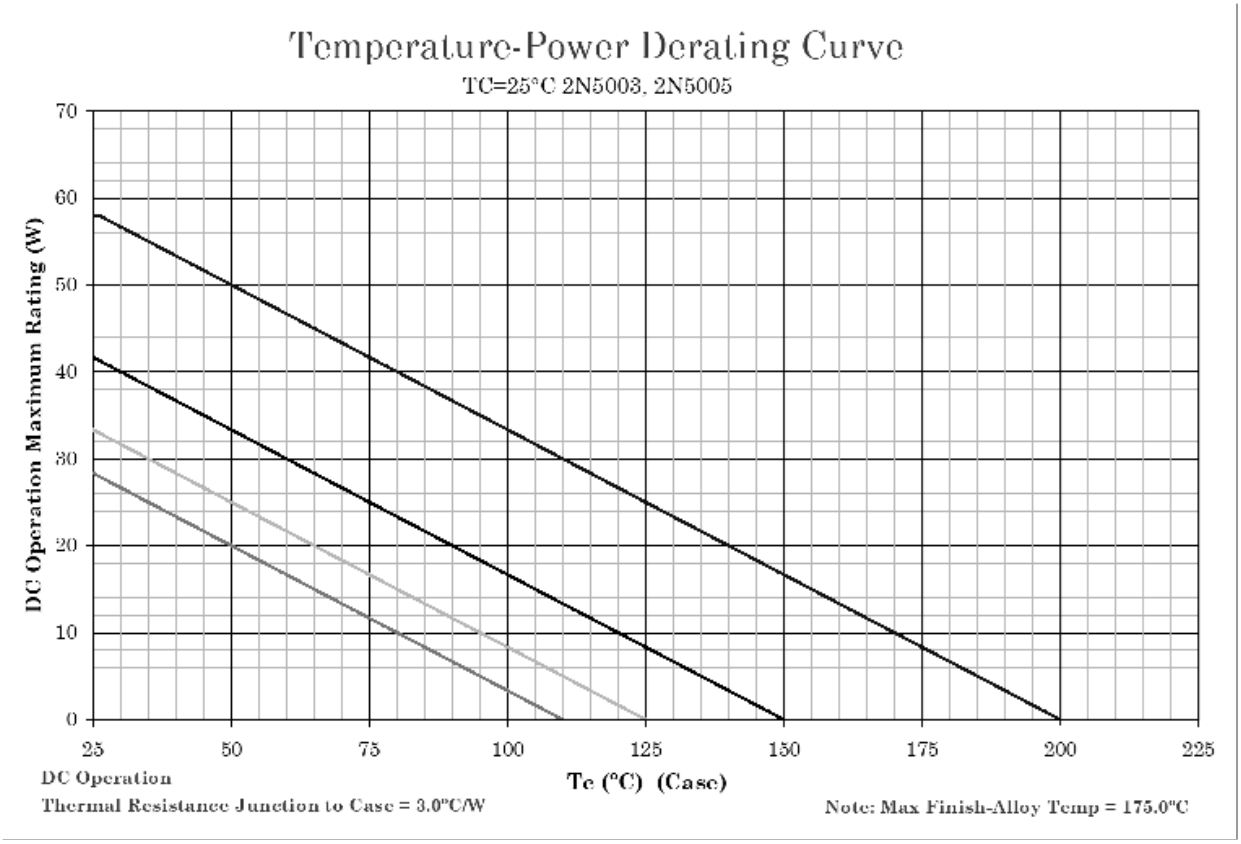
1/ For sampling plan, see MIL-PRF-19500.
2/ For resubmission of failed in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ Required only for laser marked devices with corrosion prone base metals.
7/ This hermetic seal test is an end-point to temperature-cycling in addition to electrical measurements.
8/ For end-point measurements, this test required for the following subgroups:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, step 1 of 4.4.2.2 herein (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1 and 2.

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TABLE II. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 and 4.7 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = -10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100°C	
Electrical measurements		See table I, subgroup 2 and 4.7 herein	
Subgroup 4			
Thermal impedance curves		See table E-IX of MIL-PRF-19500, subgroup 4	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B	

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NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ +200°C) and power rating specified (see 1.3 herein).
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ +125°C, and +110°C to show power rating where most users want to limit T_J in their application.

FIGURE 3. Temperature-power derating for ($R_{\theta JC}$) 2N5003 and 2N5005 (similar to TO-59).

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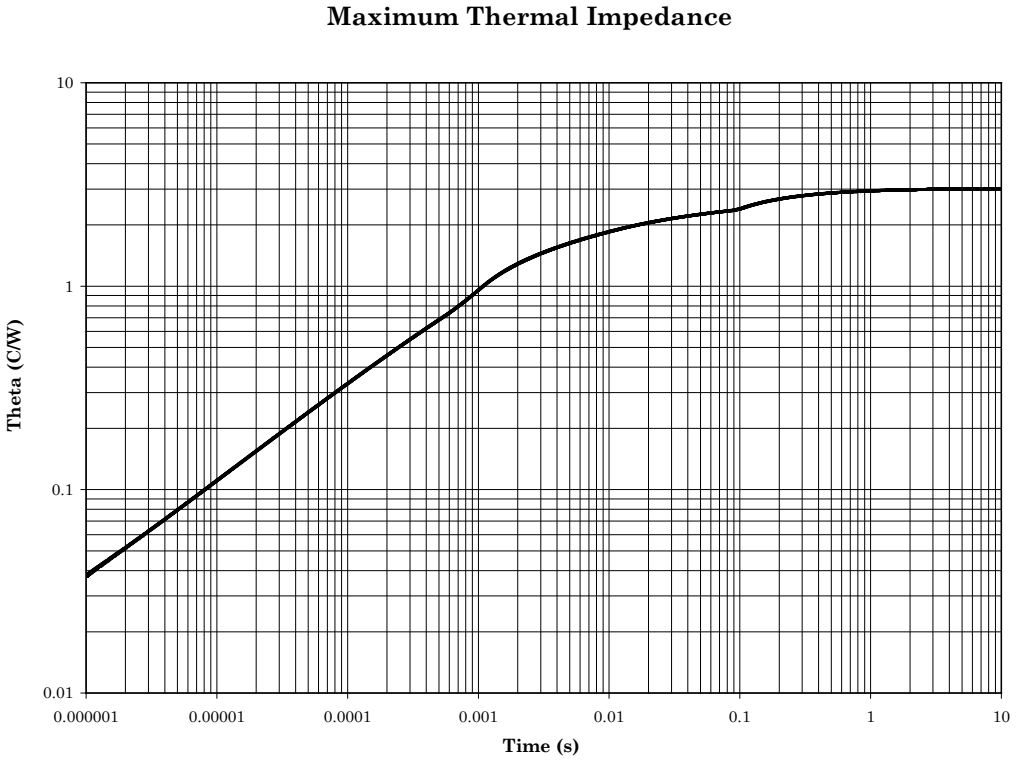
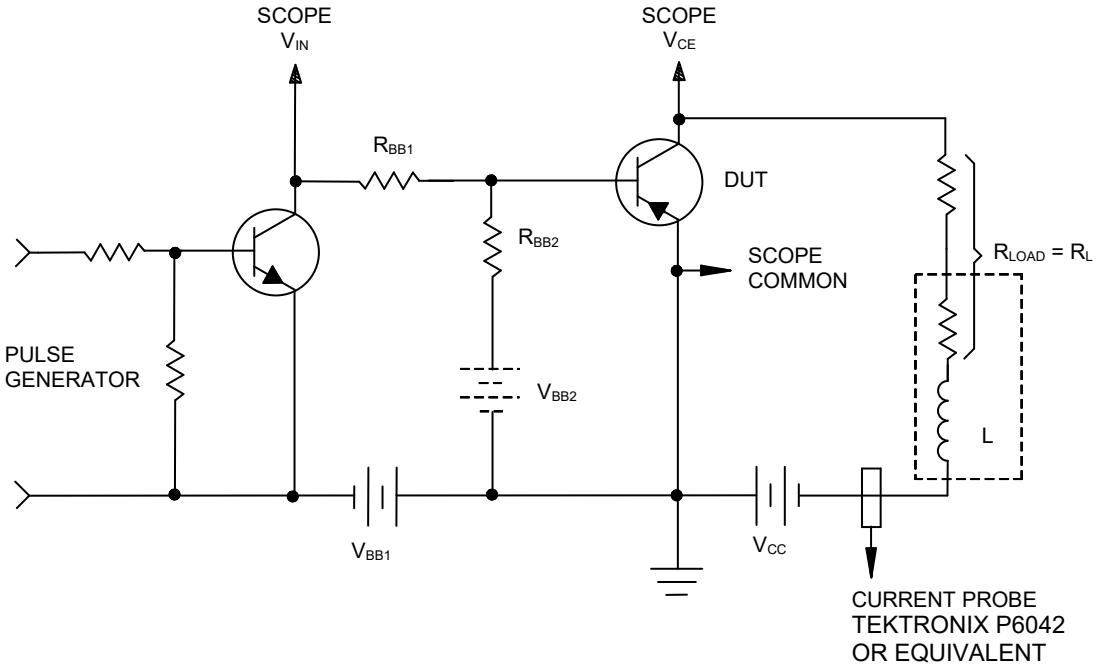


FIGURE 4. Thermal impedance graph ($R_{\theta JC}$) for 2N5003 and 2N5005 (similar to TO-59).

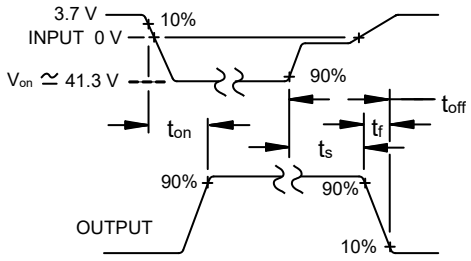
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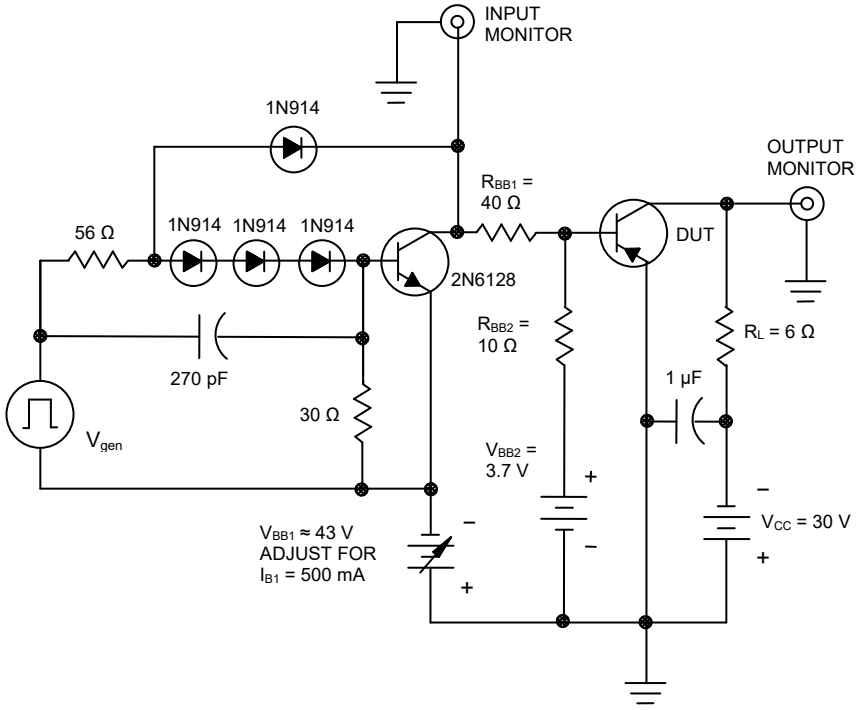


R_{BB1} = 10 Ω	V_{CC} = -10 V dc
R_{BB2} = 100 Ω	I_C = -10 A
L = 0.3 mH	V_{BB1} = -10 V dc
R_L = 0.1 Ω	V_{BB2} = -4 V dc

FIGURE 5. Unclamped inductive load energy test circuit.

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NOTES:

1. V_{gen} is -30 pulse (from 0 V) into a 50 ohm termination.
2. The V_{gen} waveform is supplied by a generator with the following characteristics:
   t_r ≤ 15 ns, t_f = 15 ns,
   Z_{out} = 50 ohm, duty cycle ≤ 2 percent.
3. Waveforms are monitored on an oscilloscope with the following characteristics:
   t_r ≤ 1 ns, R_{IN} ≥ 10 MΩ,
   C_{IN} ≤ 11.5 pF.
4. Resistors shall be non-inductive types.
5. The dc power supplies may require additional bypassing in order to minimize ringing.
6. An equivalent drive circuit may be used.

FIGURE 6. Switching time test circuit.

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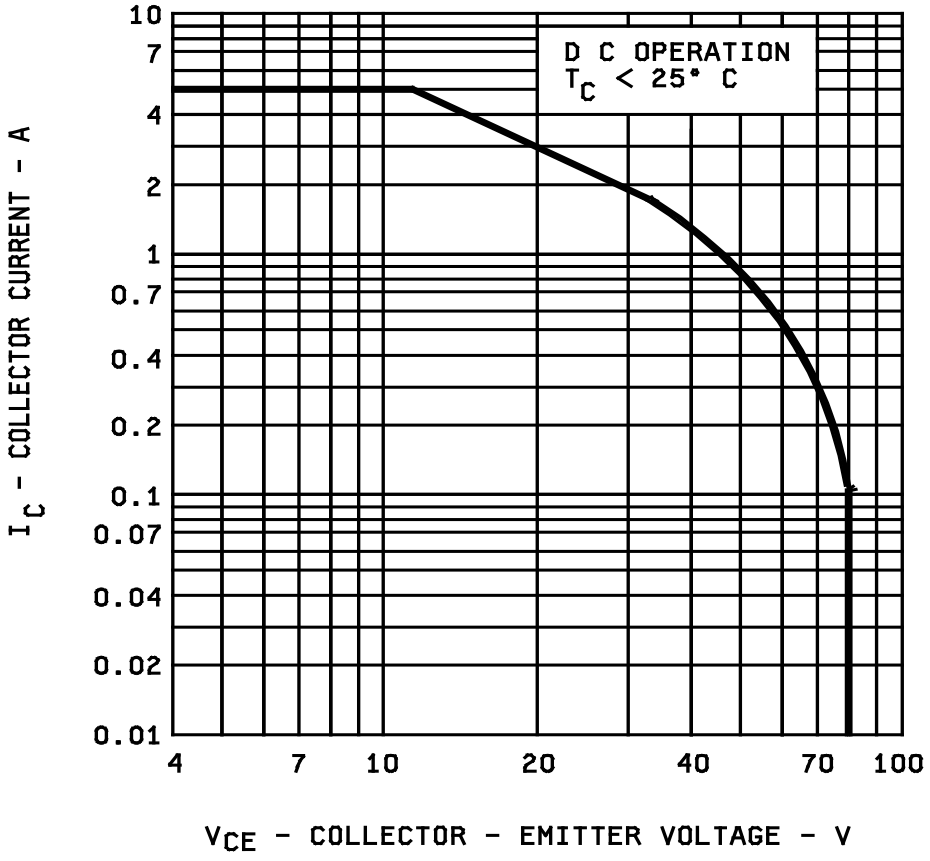


FIGURE 7. Maximum safe operating area.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

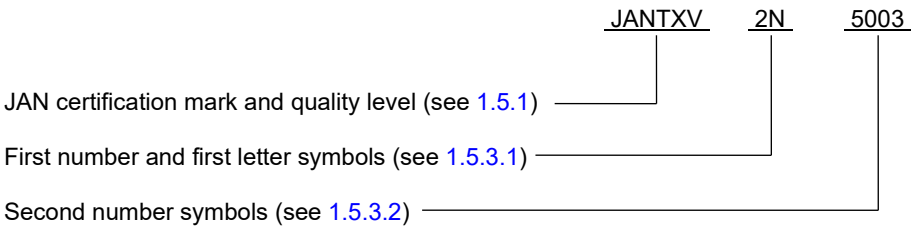
c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

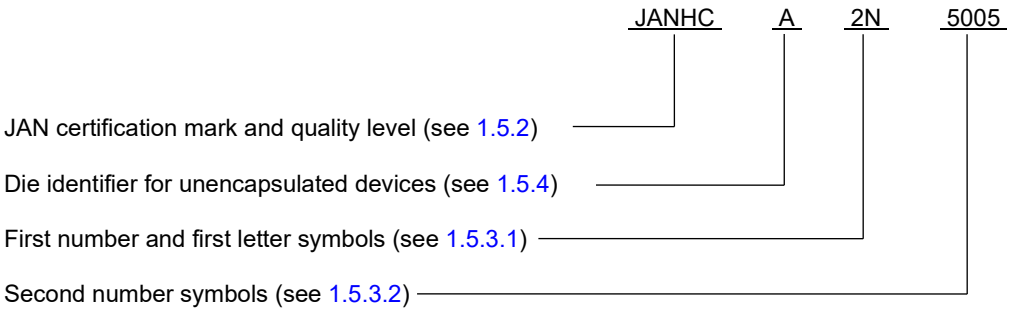
6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N5003	JANTX2N5003	JANTXV2N5003	JANS2N5003
JAN2N5005	JANTX2N5005	JANTXV2N5005	JANS2N5005

6.5.2 List of PINs for unencapsulated die. The following is a list of possible PINs for unencapsulated die available on this specification sheet.

Quality level HC	Quality level KC
JANHCA2N5003	JANKCA2N5003
JANHCA2N5005	JANKCA2N5005

6.6 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC die suppliers with the applicable letter version (e.g., JANHCA2N5003) will be identified on the QML.

JANHC ordering information	
PIN	Manufacturer CAGE
	33178
2N5003	JANHCA2N5003
2N5005	JANHCA2N5005
2N5003	JANKCA2N5003
2N5005	JANKCA2N5005

6.7 Complementary use. The devices specified herein are designed for complementary use with the 2N5002 and 2N5004 of MIL-PRF-19500/534.

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6.8 Interchangeability information. An alternate package configuration for the devices covered by this specification sheet is available on MIL-PRF-19500/612. MIL-PRF-19500/612 covers device type 2N7372, which is a TO-254 package version of the device types covered by this specification sheet. The military 2N7372 contains the same die as the military 2N5003 and 2N5005.

6.9 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

* 6.10 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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